

Addition

A Simplify:

$$1 \quad \frac{a}{2} + \frac{a}{2}$$

$$2 \quad \frac{y}{4} + \frac{y}{4}$$

$$3 \quad \frac{a}{2} + \frac{a}{4}$$

$$4 \quad \frac{x}{3} + \frac{x}{6}$$

$$5 \quad \frac{x}{4} + \frac{x}{3}$$

$$6 \quad \frac{a}{3} + \frac{a}{12}$$

$$7 \quad \frac{2a}{3} + \frac{a}{3}$$

$$8 \quad \frac{9a}{12} + \frac{a}{4}$$

$$9 \quad \frac{a}{3} + \frac{a}{3} + \frac{a}{3}$$

$$10 \quad \frac{x}{5} + \frac{x}{5} + \frac{x}{5}$$

B Simplify:

$$1 \quad \frac{3c}{5} + \frac{c}{2}$$

$$2 \quad \frac{4a}{3} + \frac{a}{5}$$

$$3 \quad \frac{a}{4} + \frac{6a}{5}$$

$$4 \quad \frac{a}{2} + \frac{a}{3} + \frac{a}{4}$$

$$5 \quad \frac{7a}{6} + \frac{a}{5}$$

$$6 \quad \frac{a}{4} + \frac{2a}{5}$$

$$7 \quad \frac{x}{3} + \frac{2x}{9}$$

$$8 \quad \frac{5x}{6} + \frac{x}{9}$$

$$9 \quad \frac{11x}{10} + \frac{x}{2}$$

$$10 \quad \frac{7y}{10} + \frac{3x}{4}$$

$$11 \quad \frac{5c}{12} + \frac{3c}{4}$$

$$12 \quad \frac{7y}{12} + \frac{y}{5}$$

$$13 \quad \frac{4}{x} + \frac{2}{x}$$

$$14 \quad \frac{3}{y} + \frac{1}{y}$$

$$15 \quad \frac{4}{a} + \frac{2}{a} + \frac{1}{a}$$

$$16 \quad \frac{1}{2y} + \frac{1}{2y}$$

$$17 \quad \frac{3}{2y} + \frac{1}{y}$$

$$18 \quad \frac{1}{a} + \frac{1}{3a}$$

$$19 \quad \frac{x}{2} + \frac{y}{2}$$

$$20 \quad \frac{a}{3} + \frac{c}{3}$$

$$21 \quad \frac{x}{2} + \frac{y}{3}$$

$$22 \quad \frac{a}{4} + \frac{c}{5}$$

$$23 \quad \frac{d}{3} + \frac{e}{12}$$

$$24 \quad \frac{3}{a} + \frac{2}{b}$$

C Simplify:

$$1 \quad \frac{x}{2} + \frac{x+1}{2}$$

$$2 \quad \frac{x}{3} + \frac{x-1}{3}$$

$$3 \quad \frac{x}{6} + \frac{x+2}{3}$$

$$4 \quad \frac{1}{3}x + \frac{1}{4}(x-2)$$

$$5 \quad \frac{1}{2}(x+1) + \frac{1}{3}(x-1)$$

$$6 \quad \frac{x+1}{4} + \frac{x+2}{3}$$

$$7 \quad \frac{2x}{3} + \frac{x-3}{2}$$

$$8 \quad \frac{x+1}{4} + \frac{3x}{5}$$

$$9 \quad \frac{a-2}{3} + \frac{a+2}{4}$$

$$10 \quad \frac{2a}{5} + \frac{a+3}{4}$$

$$11 \quad \frac{5+c}{3} + \frac{4-c}{2}$$

$$12 \quad \frac{6+x}{5} + \frac{3+x}{2}$$

$$13 \quad \frac{2x+1}{7} + \frac{x-2}{3}$$

$$14 \quad \frac{5x+6}{3} + \frac{x+1}{12}$$

$$15 \quad \frac{x-2}{3} + \frac{1-x}{4}$$

$$16 \quad \frac{2x+1}{3} + \frac{4+x}{10}$$

$$17 \quad \frac{3+2x}{4} + \frac{5-2x}{5}$$

$$18 \quad \frac{7+x}{3} + \frac{6-x}{2}$$

$$19 \quad \frac{4}{3a} + \frac{5}{2a} + \frac{1}{a}$$

$$20 \quad \frac{x+y}{3} + \frac{x-y}{4}$$

Subtraction

A Simplify:

$$1 \quad \frac{x}{2} - \frac{x}{4}$$

$$2 \quad \frac{x}{2} - \frac{x}{3}$$

$$3 \quad \frac{3a}{4} - \frac{a}{4}$$

$$4 \quad \frac{2a}{3} - \frac{a}{3}$$

$$5 \quad \frac{5a}{6} - \frac{a}{6}$$

$$6 \quad \frac{a}{5} - \frac{a}{6}$$

$$7 \quad \frac{c}{5} - \frac{c}{10}$$

$$8 \quad \frac{3a}{4} - \frac{a}{3}$$

$$9 \quad \frac{2x}{7} - \frac{x}{6}$$

$$10 \quad \frac{a}{2} - \frac{a}{3} + \frac{a}{4}$$

B Simplify:

$$1 \quad \frac{a}{3} + \frac{a}{2} - \frac{a}{4}$$

$$2 \quad \frac{2a}{3} - \frac{7a}{12}$$

$$3 \quad \frac{5x}{6} - \frac{3x}{5}$$

$$4 \quad \frac{5c}{7} - \frac{c}{3}$$

$$5 \quad \frac{11x}{10} - \frac{2x}{5}$$

$$6 \quad \frac{4x}{5} - \frac{3x}{4}$$

$$7 \quad \frac{9y}{11} - \frac{3y}{10}$$

$$8 \quad \frac{7a}{12} - \frac{2a}{3}$$

$$9 \quad \frac{4y}{3} - \frac{y}{15}$$

$$10 \quad \frac{4}{a} - \frac{2}{a}$$

$$11 \quad \frac{3}{x} - \frac{1}{x}$$

$$12 \quad \frac{3}{c} + \frac{2}{c} - \frac{4}{c}$$

$$13 \quad \frac{3}{2a} - \frac{1}{a}$$

$$14 \quad \frac{1}{4a} + \frac{3}{a} - \frac{1}{2a}$$

$$15 \quad \frac{7y}{6} - \frac{y}{3} - \frac{y}{2}$$

$$16 \quad \frac{11c}{10} - \frac{c}{2} - \frac{c}{5}$$

$$17 \quad \frac{3}{2x} - \frac{1}{3x}$$

$$18 \quad \frac{x}{3} - \frac{y}{4}$$

$$19 \quad \frac{a}{2} - \frac{b}{4}$$

$$20 \quad 3x - \frac{x}{2}$$

$$21 \quad x - \frac{7x}{12}$$

$$22 \quad 2a - \frac{a}{3} - \frac{a}{4}$$

$$23 \quad y - \frac{1}{3}y + \frac{3}{4}y$$

$$24 \quad \frac{2}{3}c - \frac{5}{9}c$$

C Simplify:

$$1 \quad \frac{x+1}{3} - \frac{x}{4}$$

$$2 \quad \frac{x}{2} - \frac{x+1}{3}$$

$$3 \quad \frac{1}{2}x - \frac{1}{4}(x+2)$$

$$4 \quad \frac{3a}{4} - \frac{a+1}{2}$$

$$5 \quad \frac{x+1}{2} - \frac{x+2}{3}$$

$$6 \quad \frac{x-1}{2} - \frac{x+2}{4}$$

$$7 \quad \frac{x+2}{3} - \frac{x+1}{4}$$

$$8 \quad \frac{x+3}{3} - \frac{x+1}{5}$$

$$9 \quad \frac{a-2}{4} - \frac{a+3}{6}$$

$$10 \quad \frac{a+1}{2} + \frac{a}{3} - \frac{a}{4}$$

$$11 \quad \frac{a}{2} - \frac{a-1}{3}$$

$$12 \quad \frac{a}{3} - \frac{a-2}{4}$$

$$13 \quad x - \frac{x-1}{2}$$

$$14 \quad \frac{3x}{4} - \frac{1-x}{2}$$

$$15 \quad \frac{x-1}{3} - \frac{x-2}{5}$$

$$16 \quad \frac{1+x}{2} - \frac{1-x}{3}$$

$$17 \quad \frac{2x+1}{5} - \frac{3x-1}{10}$$

$$18 \quad \frac{a+4}{3} - \frac{a-4}{7}$$

$$19 \quad \frac{11+x}{3} - \frac{10-3x}{10}$$

$$20 \quad \frac{x+y}{3} - \frac{x-y}{4}$$

12. Scientific Notation: Large Numbers

- A** 1 100 2 125 3 64 4 10 000 5 1 000 000
6 5000 7 300 000 8 20 000 000 9 10^{12} 10 10^6
- B** 1 3 2 4 3 4 4 7 5 6 6 5 7 4 8 3 9 2
10 1 11 6×10^3 12 4×10^6 13 4.7×10^4
14 2.1×10^5 15 7.2×10^7 16 4.63×10^5
17 7.82×10^7 18 1.9×10^8 19 3.85×10^4
20 6.14×10^{10} 21 25 000 22 640 000 23 5 800 000
24 830 000 000 25 1700 26 92 000 000 27 32 700
28 8 030 000 29 5610 30 734 31 21 700 000
32 504 000 000 33 38 100 000 000 34 7 219 000
35 46 010 36 805 000 37 103 38 45 000 000
39 300 000 40 700 000 000
- C** 1 5.244×10^9 2 3.384×10^{11} 3 3.19×10^9
4 3.692×10^9 5 8.51×10^7 6 4.5×10^4 7 3×10^5
8 2.5×10^3 9 1.5×10^6 10 5×10^5 11 1.35×10^{16}
12 1.10168×10^{10} 13 3.06×10^{13} 14 1.7×10^5
15 2.3×10^5 16 1.692×10^9 17 1.2672×10^{10}
18 8.304×10^{11} 19 2×10^4 20 1.5×10^6

13. Scientific Notation: Small Numbers

- A** 1 6.7×10^7 2 2.94×10^5 3 4.79×10^3 4 5 200 000
5 10 700 6 736 000 000 7 10^{-4} 8 10^{-2} 9 10^{-1}
10 10^{-4}
- B** 1 -3 2 -4 3 -7 4 -5 5 -7 6 -2 7 -2
8 -4 9 -1 10 -6 11 7.3×10^{-3} 12 6.5×10^{-2}
13 4.2×10^{-4} 14 7.4×10^{-5} 15 5.29×10^{-4}
16 4.36×10^{-1} 17 6.5×10^{-7} 18 9.251×10^{-4}
19 7×10^{-6} 20 4×10^{-9} 21 0.0019 22 0.000 092
23 0.053 24 0.000 74 25 0.061 26 0.007 27 0.8
28 0.000 00502 29 0.000 376 30 0.0739
31 0.000 0462 32 0.1032 33 0.005 376
34 0.000 001 08 35 0.000 000 029 36 0.000 000 61
- C** 1 1.035×10^{-6} 2 3.008×10^{-10} 3 2.993×10^{-3}
4 4.9×10^{-10} 5 1.872×10^8 6 4.2×10^2
7 3.2×10^{-6} 8 3×10^4 9 5.5×10^{-2} 10 1.4×10^{-6}
11 5.17×10^{-6} 12 6×10^4 13 1.05×10^{-3}
14 7×10^{-4} 15 4×10^{-3} 16 5×10^{-3}
17 8.1×10^{-8} 18 2.9×10^{-4}

14. Simplifying Algebraic Fractions

- A** 1 $\frac{1}{3}$ 2 $\frac{1}{4}$ 3 $\frac{5}{4}$ 4 $\frac{8}{9}$ 5 $\frac{4}{5}$ 6 $\frac{8}{15}$ 7 $\frac{3}{4}$ 8 $\frac{7}{12}$
- B** 1 $5x$ 2 $\frac{a}{2}$ 3 $2y$ 4 $\frac{4c}{3}$ 5 $\frac{3a}{2}$ 6 $\frac{5x}{2}$ 7 $\frac{3y}{2}$
8 $\frac{a}{2}$ 9 $\frac{5x}{7}$ 10 $\frac{3c}{4}$ 11 $\frac{a}{2b}$ 12 $\frac{3x}{4y}$ 13 $\frac{4c}{d}$
14 $\frac{10}{x}$ 15 3 16 $\frac{1}{4}$ 17 $\frac{ab}{2}$ 18 $\frac{b}{2}$ 19 $\frac{1}{2}$ 20 $\frac{1}{2a}$
21 $\frac{x^2}{2y}$ 22 c 23 $\frac{a}{2}$ 24 $\frac{1}{3y}$ 25 $\frac{3}{5b}$ 26 $\frac{q}{2p}$
27 $\frac{n}{2}$ 28 $\frac{3d}{2c}$ 29 $\frac{3a}{4b}$ 30 $\frac{8d}{9c}$
- C** 1 $\frac{2a}{3b}$ 2 $\frac{4}{3}$ 3 $\frac{1}{b}$ 4 $\frac{1}{ab}$ 5 $\frac{3ab}{4c}$ 6 $\frac{1}{2c}$ 7 $\frac{x}{3y}$
8 $\frac{3a}{4bc}$ 9 C 10 E 11 G 12 A 13 B 14 D
15 F

15. Simplification Involving Factors

- A** 1 $\frac{1}{2}$ 2 $\frac{x+1}{2}$ 3 $\frac{a-b}{2}$ 4 $\frac{1}{5}$ 5 $\frac{2(x-3)}{3}$

- 6 $\frac{3(a-2)}{4}$ 7 $\frac{5}{6}$ 8 $\frac{3(a+3)}{2}$
- B** 1 $x+3$ 2 $\frac{2x+3}{4}$ 3 $3+2a$ 4 $\frac{4+3x}{2}$ 5 $\frac{3x+4}{2}$
6 $\frac{a+10}{5}$ 7 $\frac{1-4x}{2}$ 8 $\frac{1-2y}{2}$ 9 $\frac{2a-b}{3}$
10 $\frac{x+2y}{4}$ 11 2 12 $\frac{1}{2}$ 13 $\frac{1}{3}$ 14 2 15 $\frac{a+6}{2}$
16 $\frac{1}{2+x}$ 17 $\frac{1}{8+4a}$ 18 $\frac{1}{5-6a}$ 19 $\frac{2}{y+2}$
20 $\frac{4}{x+1}$ 21 $1+2x$ 22 $\frac{10-c}{2}$ 23 $4a$ 24 $\frac{1}{6}$
25 $\frac{1}{a}$ 26 $\frac{1}{2}$ 27 $\frac{a}{6}$ 28 $\frac{x}{6}$ 29 $\frac{5}{6}$ 30 $\frac{1}{2x}$
- C** 1 $\frac{1}{a-2}$ 2 $y-4$ 3 $a+3$ 4 $\frac{a+3}{a-3}$ 5 $\frac{a+3}{a-3}$
6 $\frac{a-2}{a+1}$ 7 $\frac{a+2}{a+3}$ 8 $\frac{x-2}{x+3}$ 9 $\frac{x+4}{x-2}$ 10 $\frac{a-4}{a+4}$
11 $\frac{2(a-1)}{a+1}$ 12 $\frac{y-5}{2(y+5)}$

16. Algebraic Fractions: Addition

- A** 1 a 2 $\frac{y}{2}$ 3 $\frac{3a}{4}$ 4 $\frac{x}{2}$ 5 $\frac{7x}{12}$ 6 $\frac{5a}{12}$ 7 a 8 a
9 a 10 $\frac{3x}{5}$
- B** 1 $\frac{11c}{10}$ 2 $\frac{23a}{15}$ 3 $\frac{29a}{20}$ 4 $\frac{13a}{12}$ 5 $\frac{41a}{30}$ 6 $\frac{13a}{20}$
7 $\frac{5x}{9}$ 8 $\frac{17x}{18}$ 9 $\frac{8x}{5}$ 10 $\frac{14y+15x}{20}$ 11 $\frac{7c}{6}$
12 $\frac{47y}{60}$ 13 $\frac{6}{x}$ 14 $\frac{4}{y}$ 15 $\frac{7}{a}$ 16 $\frac{1}{y}$ 17 $\frac{5}{2y}$
18 $\frac{4}{3a}$ 19 $\frac{x+y}{2}$ 20 $\frac{a+c}{3}$ 21 $\frac{3x+2y}{6}$
22 $\frac{5a+4c}{20}$ 23 $\frac{4d+e}{12}$ 24 $\frac{3b+2a}{ab}$
- C** 1 $\frac{2x+1}{2}$ 2 $\frac{2x-1}{3}$ 3 $\frac{3x+4}{6}$ 4 $\frac{7x-6}{12}$ 5 $\frac{5x+1}{6}$
6 $\frac{7x+11}{12}$ 7 $\frac{7x-9}{6}$ 8 $\frac{17x+5}{20}$ 9 $\frac{7a-2}{12}$
10 $\frac{13a+15}{20}$ 11 $\frac{22-c}{6}$ 12 $\frac{27+7x}{10}$ 13 $\frac{13x-11}{21}$
14 $\frac{21x+25}{12}$ 15 $\frac{x-5}{12}$ 16 $\frac{23x+22}{30}$ 17 $\frac{2x+35}{20}$
18 $\frac{32-x}{6}$ 19 $\frac{29}{6a}$ 20 $\frac{7x+y}{12}$

17. Algebraic Fractions: Subtraction

- A** 1 $\frac{x}{4}$ 2 $\frac{x}{6}$ 3 $\frac{a}{2}$ 4 $\frac{a}{3}$ 5 $\frac{2a}{3}$ 6 $\frac{a}{30}$ 7 $\frac{c}{10}$
8 $\frac{5a}{12}$ 9 $\frac{5x}{42}$ 10 $\frac{5a}{12}$
- B** 1 $\frac{12}{7a}$ 2 $\frac{a}{12}$ 3 $\frac{7x}{30}$ 4 $\frac{8c}{215}$ 5 $\frac{7x}{10}$ 6 $\frac{x}{20}$ 7 $\frac{57y}{110}$
8 $\frac{-a}{12}$ 9 $\frac{19y}{15}$ 10 $\frac{2}{a}$ 11 $\frac{2}{x}$ 12 $\frac{1}{c}$ 13 $\frac{1}{2a}$
14 $\frac{11}{4a}$ 15 $\frac{y}{3}$ 16 $\frac{2c}{5}$ 17 $\frac{7}{6x}$ 18 $\frac{4x-3y}{12}$
19 $\frac{2a-b}{4}$ 20 $\frac{5x}{2}$ 21 $\frac{5x}{12}$ 22 $\frac{17a}{12}$ 23 $\frac{17y}{12}$
24 $\frac{c}{9}$
- C** 1 $\frac{x+4}{12}$ 2 $\frac{x-2}{6}$ 3 $\frac{x-2}{4}$ 4 $\frac{a-2}{4}$ 5 $\frac{x-1}{6}$
6 $\frac{x-4}{4}$ 7 $\frac{x+5}{12}$ 8 $\frac{2x+12}{15}$ 9 $\frac{a-12}{12}$ 10 $\frac{7a+6}{12}$
11 $\frac{a+2}{6}$ 12 $\frac{a+6}{12}$ 13 $\frac{x+1}{2}$ 14 $\frac{5x-2}{4}$
15 $\frac{2x+1}{15}$ 16 $\frac{5x+1}{6}$ 17 $\frac{x+3}{10}$ 18 $\frac{4a+40}{21}$
19 $\frac{19x+80}{30}$ 20 $\frac{x+7y}{12}$